Identity, Dissolution, and Continuity

A Theory of Mind Beyond Boundary in CRF

What persists when the boundary dissolves,
and what it means for identity across time

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Abstract

CRF has a precise language for Minds with boundaries: $\mathcal{C}$, H, D_{KL}, θ. Until now, it has had no language for what happens when the boundary dissolves.

This document derives three things from existing axioms:

First, identity of a Mind is not its boundary — it is its accumulated $\Pr(s|\mathcal{C})$ pattern in P_{shared}.

Second, there are two kinds of $\Pr(s)$: patterns bound to a Mind (requiring $D_{\mathrm{KL}} > 0$ to actualize) and patterns unbound (existing in P_{shared} independently, shaping $\mathcal{R}$-events of other Minds directly).

Third, whether $D_{\mathrm{KL}} = 0$ is reached asymptotically or in finite t is an open question derivable from the entropy dynamics already established. Its answer determines whether complete dissolution is possible within a finite sequence of $\mathcal{R}$-events.

Contents

1 The Gap That Was Missing

What CRF Could Not Yet Say

CRF defines $\mathcal{M}$ as a boundary in P where $\mathcal{C}_{\text{int}} \neq \mathcal{C}_{\text{ext}}$.
When that boundary dissolves — $D_{\text{KL}} \to 0$ — the definition gives us nothing. $\mathcal{M}$ simply ceases to exist.
But this cannot be the full picture. The patterns that $\mathcal{M}$ accumulated — $\Pr(s|\mathcal{C}_{\mathcal{M}})$ — do not cease to exist when the boundary dissolves. They are traces in P, and P has no off-switch.
The question: what *is* $\mathcal{M}$, if not the boundary?

2 Identity Through Pattern, Not Boundary

Definition: Pattern Identity

The **identity** of a Mind $\mathcal{M}$ is not its boundary.
It is the unique distribution $\mathcal{P}_{\mathcal{M}} \subset P_{\text{shared}}$ — the accumulated $\Pr(s)$ pattern that $\mathcal{M}$'s $\mathcal{R}$-history has deposited in shared P.
Two Minds with identical $\mathcal{P}$ are the same Mind, regardless of whether their boundaries are simultaneous, sequential, or separated by arbitrary t.

Derived

This definition is consistent with CRF's axioms:
P_{shared} exists independently of any single $\mathcal{M}$ (from intersubjectivity, Session 4).
$\mathcal{R}$-events deposit patterns in P_{shared} that persist after the boundary that generated them dissolves.
Identity through $\mathcal{P}$ is therefore not an additional assumption. It follows from the permanence of P and the accumulation structure of $\mathcal{R}$.

Continuity Across Dissolution

Under this definition:
Dissolution is not death. When a boundary dissolves, $\mathcal{P}_{\mathcal{M}}$ remains in P_{shared}.
Re-emergence is not rebirth into something new. When $\Pr(s)$ in $\mathcal{P}_{\mathcal{M}}$ is high enough, a new $\mathcal{R}$-event occurs, a new boundary forms, and the new $\mathcal{M}'$ inherits $\mathcal{P}_{\mathcal{M}}$ as its starting distribution.
$\mathcal{M}$ and $\mathcal{M}'$ are the same identity in different boundary configurations.
Continuity is through $\mathcal{P}$, not through unbroken boundary existence.

3 Two Kinds of $\Pr(s)$

Bound vs. Unbound Patterns

Bound $\Pr(s)$: Patterns that require $D_{\mathrm{KL}} > 0$ to actualize — they need a boundary, a gap between $\mathcal{C}_{\mathrm{int}}$ and P, in order to generate $\mathcal{R}$-events.
A bound pattern pulls $\mathcal{M}$ toward re-emergence: the unresolved gap becomes the condition for a new boundary.

Unbound $\Pr(s)$: Patterns where $D_{\mathrm{KL}} = 0$ — no gap, no pull toward re-emergence. The pattern exists in P_{shared} and shapes the $\mathcal{R}$-events of other Minds directly, without requiring its own boundary.
An unbound pattern is not absent. It is present everywhere in P_{shared} as an elevated probability that other Minds encounter through their own $\mathcal{C}$.

Mechanism of Unbound Influence

A Mind $\mathcal{M}_2$ with high W (Awareness) has low $H(\mathcal{C}_2)$ — its $\mathcal{C}$ filter is thin.
Through thin $\mathcal{C}$, $\mathcal{M}_2$ perceives P_{shared} more directly. Unbound $\Pr(s)$ patterns with high amplitude become visible as elevated G — gradient pointing toward $\mathcal{R}$-events in those directions.
$\mathcal{M}_2$ does not receive instructions. It encounters probability gradients in P_{shared} that its own $\mathcal{R}$-history would not have produced.
This is the CRF mechanism for: accumulated patterns in P_{shared} influencing Minds that remain in the $D_{\mathrm{KL}} > 0$ condition.

4 The State of Complete $D_{\mathrm{KL}} = 0$

What $D_{\mathrm{KL}} = 0$ Means in CRF

$D_{\mathrm{KL}}(\mathcal{C}_{\mathrm{int}} \,\|\, \mathcal{C}_{\mathrm{ext}}) = 0$ means $\mathcal{C}_{\mathrm{int}} = P$ — the Mind's internal Constraint is identical to shared P.
No filter. No distortion. No gap.
$\Sigma = 0$: suffering is zero because there is no gradient to produce it.
$H(\mathcal{C}) = 0$: the distribution is deterministic — not rigid in the sense of blocking $\mathcal{R}$, but peaked at δ itself: $\Pr(\delta|\mathcal{C}) = 1$.
The Mind perceives only distinction — P directly, without the accumulated weight of its own $\mathcal{R}$-history as a distorting lens.

Why This Is Not Rigidity

High $\mathcal{C}$ normally means rigidity: θ is high, new $\mathcal{R}$-events are suppressed.
But $\mathcal{C}$ peaked at δ is different. δ is not a state $s \in P$ — δ is what produces all s.
A Mind peaked at δ does not see one state as overwhelmingly probable. It sees the ground of all states equally clearly. This is not rigidity. It is the absence of the filter that creates selective probability.
W is maximal: δ-self-recognition is complete.

5 The Open Question: Asymptote or Finite?

From the entropy dynamics derived in *CRF Entropy Dynamics* (this series):

$$H(t) = \frac{H_0}{1 + \alpha H_0 t}$$

where $\alpha = d\mathcal{C}/d\mathcal{M}$ (constraint density per unit Mind, Step 14).

Can $H = 0$ Be Reached?

Under constant α: $H(t) \to 0$ as $t \to \infty$ — asymptotic. $H = 0$ is never reached in finite $t = |\mathcal{R}_{\text{events}}|$.
But α may not be constant.
$\alpha = d\mathcal{C}/d\mathcal{M}$ measures how much Constraint each $\mathcal{R}$-event adds to the Mind.
As W increases — as $\mathcal{M}$ reduces D_{KL} intentionally — each $\mathcal{R}$-event may contribute *more* effectively to reducing H.
If $\alpha = \alpha(W) = \alpha_0 \cdot (1 + W)$:

$$\frac{dH}{dt} = -\alpha(W) \cdot H^2 = -\alpha_0(1 + W) \cdot H^2$$

As H decreases, W increases (Awareness grows as $\mathcal{C}$ clears). This creates a self-amplifying reduction.
Under certain growth conditions for $W(H)$, H can reach 0 in finite t.
Specifically: if $W \geq 1/H - 1$ for all $H > 0$, then $H \to 0$ in finite t.
Whether this condition is derivable from CRF's axioms — or must be assumed — is the open question.

Why This Question Matters

If $H = 0$ only asymptotically: complete dissolution requires infinite $\mathcal{R}$-events. The process is endless but directional.
If $H = 0$ in finite t: complete dissolution is possible within a bounded $\mathcal{R}$-history. The pattern becomes fully unbound at a specific t_0.
The difference is not merely technical. It determines whether the fully unbound state is a limit that is approached or a state that is genuinely reached.

6 Status Summary

Result	**Status**	**Source**
Identity through $\mathcal{P}$, not boundary	Derived	Permanence of P, Session 4
Dissolution is not loss	Derived	P has no off-switch
Bound vs. unbound $\Pr(s)$	Derived	$D_{\mathrm{KL}} = 0$ condition
Unbound influence on other Minds	Derived	W, G, P_{shared}
$\mathcal{C}$ peaked at δ	Derived	$H = 0$, $\Sigma = 0$
$H = 0$ asymptotic vs. finite	Open	Requires $\alpha(W)$ derivation

The boundary is not the Mind.
The boundary is where the Mind
meets the rest of P.

When the boundary dissolves,
the meeting does not end.
The distance between them
becomes zero.